AP Calculus BC Unit 7 Review Book

Differential Equations

Detailed, beginner-friendly, and complete final review

Key idea. This unit is about connecting *rates of change* to *functions*. A differential equation tells you how a quantity changes. Your job is to model, verify, sketch or interpret slope fields, approximate values with Euler's Method, solve separable equations, apply initial conditions, and recognize exponential and logistic behavior.

1. Unit 7 Big Picture

The official Unit 7 topics are:

1. Modeling situations with differential equations
2. Verifying solutions for differential equations
3. Sketching slope fields
4. Reasoning using slope fields
5. Euler's Method (BC)
6. Separation of variables
7. Initial conditions and particular solutions
8. Exponential models
9. Logistic models (BC)

A differential equation may look like any of the following:

$$\frac{dy}{dx} = f(x), \qquad \frac{dy}{dx} = g(y), \qquad \frac{dy}{dx} = h(x, y).$$

When the variable is time, you often see

$$\frac{dy}{dt} = f(t, y).$$

2. 7.1 Modeling Situations with Differential Equations

Core idea

Translate the words into a rate equation. The derivative must match the quantity that is changing.

Key idea. Useful modeling templates:

$$\frac{dy}{dt} = ky \quad \text{(rate proportional to the current amount)}$$

$$\frac{dy}{dt} = k(A - y) \quad \text{(rate proportional to distance from a target value)}$$

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right) \quad \text{(logistic model)}$$

Translation patterns

Words in the prompt	Model
The rate of change of y is proportional to y.	$\frac{dy}{dt} = ky$
The growth rate of population P is proportional to the current population.	$\frac{dP}{dt} = kP$
The temperature changes at a rate proportional to the difference between room temperature A and the current temperature T.	$\frac{dT}{dt} = k(A - T)$
The slope depends only on x.	$\frac{dy}{dx} = f(x)$
The slope depends only on y.	$\frac{dy}{dx} = g(y)$
There is a carrying capacity L.	$\frac{dP}{dt} = kP(1 - P/L)$

Worked example. A bacteria population $P(t)$ grows at a rate proportional to the amount present. The constant of proportionality is 0.3. Write the model.

Because the changing quantity is P, the derivative is $\frac{dP}{dt}$. Proportional to the current amount means the right side is a constant times P:

$$\frac{dP}{dt} = 0.3P.$$

Worked example. The temperature $T(t)$ of a cup of coffee changes at a rate proportional to the difference between the room temperature 72 and the current temperature. Write a differential equation.

The difference from room temperature is $72 - T$, so the model is

$$\frac{dT}{dt} = k(72 - T).$$

Common mistake. If the prompt says "rate of change of y," the left side is $\frac{dy}{dt}$ or $\frac{dy}{dx}$. Do not write an equation with y alone on the left side.

3. 7.2 Verifying Solutions for Differential Equations

Core idea

To verify a solution, differentiate and substitute. If there is an initial condition, check that separately.

Key idea. Verification checklist:

1. Differentiate the proposed function.
2. Substitute into the differential equation.
3. Simplify both sides.
4. If an initial condition is given, check it too.

Worked example. Verify that $y = Ce^{2x}$ solves

$$\frac{dy}{dx} = 2y.$$

Differentiate:

$$\frac{dy}{dx} = 2Ce^{2x}.$$

Since $y = Ce^{2x}$, the right-hand side is

$$2y = 2Ce^{2x}.$$

Therefore, the equation is satisfied for all x.

Worked example. Determine whether $y = x^2 + 1$ solves the initial-value problem

$$\frac{dy}{dx} = 2x, \qquad y(0) = 2.$$

Differentiate:

$$\frac{dy}{dx} = 2x.$$

So the differential equation is satisfied. But

$$y(0) = 0^2 + 1 = 1 \neq 2.$$

Therefore, it solves the differential equation but not the given initial-value problem.

4. 7.3 Sketching Slope Fields

Core idea

A slope field shows the value of $\frac{dy}{dx}$ at many points.

Key idea. Quick pattern recognition:

- If $\frac{dy}{dx} = f(x)$, the slope is the same in vertical columns.
- If $\frac{dy}{dx} = g(y)$, the slope is the same in horizontal rows.
- If $\frac{dy}{dx} = 0$, the segment is horizontal.

- Positive slope means solution curves rise.
- Negative slope means solution curves fall.

Worked example. Describe the slope field for

$$\frac{dy}{dx} = x.$$

The slope depends only on x, so each vertical column has one common slope. Along $x = 0$, slopes are zero. For $x > 0$, slopes are positive. For $x < 0$, slopes are negative.

Worked example. Describe the slope field for

$$\frac{dy}{dx} = y - 2.$$

The slope depends only on y, so each horizontal row has one common slope. Along $y = 2$, the slope is zero. Above $y = 2$, slopes are positive. Below $y = 2$, slopes are negative.

Common mistake. A slope field is not one graph. It is a field of local tangent directions for many possible solution curves.

5. 7.4 Reasoning Using Slope Fields

Core idea

Even without solving the equation, you can determine whether a solution is increasing, decreasing, leveling off, or moving away from an equilibrium.

Equilibrium solutions

If $\frac{dy}{dx} = 0$ along the horizontal line $y = c$, then $y = c$ is an equilibrium solution.

Worked example. Suppose a slope field has horizontal segments all along the line $y = 3$, positive slopes above that line, and negative slopes below it. What does that mean?

It means $y = 3$ is an equilibrium solution. Because slopes are positive above and negative below, solution curves move away from $y = 3$, so this is an unstable equilibrium pattern.

Worked example. Suppose a slope field has negative slopes above $y = 5$, positive slopes below $y = 5$, and zero slope on $y = 5$. What happens to solutions near $y = 5$?

Solutions move toward $y = 5$ from both sides, so $y = 5$ is a stable equilibrium.

6. 7.5 Euler's Method (BC)

Core idea

Euler's Method uses tangent-line steps to estimate values of a solution numerically.

Key idea. If

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0,$$

and the step size is h, then

$$x_{n+1} = x_n + h,$$
$$y_{n+1} = y_n + h\, f(x_n, y_n).$$

Example 1

Use Euler's Method with $h = 0.2$ to estimate $y(0.4)$ for

$$\frac{dy}{dx} = x + y, \qquad y(0) = 1.$$

Worked example. Start at $(0, 1)$.

First step:

$$f(0, 1) = 0 + 1 = 1,$$

so

$$y_1 = 1 + 0.2(1) = 1.2, \qquad x_1 = 0.2.$$

Second step:

$$f(0.2, 1.2) = 0.2 + 1.2 = 1.4,$$

so

$$y_2 = 1.2 + 0.2(1.4) = 1.48.$$

Therefore,

$$y(0.4) \approx 1.48.$$

Example 2

Use one Euler step with $h = 0.1$ to estimate $y(1.1)$ for

$$\frac{dy}{dx} = x^2 - y, \qquad y(1) = 2.$$

Worked example. At $(1, 2)$, the slope is

$$f(1, 2) = 1^2 - 2 = -1.$$

Thus,

$$y(1.1) \approx 2 + 0.1(-1) = 1.9.$$

Common mistake. Update both x and y after each step. Many errors happen because students keep using the original point instead of the new point.

7. 7.6 General Solutions by Separation of Variables

Core idea

A differential equation is separable if you can move all the y-expressions to one side and all the x-expressions to the other side.

Key idea. If

$$\frac{dy}{dx} = g(x)h(y),$$

then you may rewrite it as

$$\frac{1}{h(y)}\,dy = g(x)\,dx$$

and integrate both sides.

Worked example. Solve

$$\frac{dy}{dx} = 3xy.$$

Separate variables:

$$\frac{1}{y}\,dy = 3x\,dx.$$

Integrate:

$$\int \frac{1}{y}\,dy = \int 3x\,dx$$

$$\ln|y| = \frac{3x^2}{2} + C.$$

Exponentiate:

$$y = Ce^{\frac{3x^2}{2}}.$$

Worked example. Solve

$$\frac{dy}{dx} = \frac{x^2}{1+y^2}.$$

Separate variables:

$$(1+y^2)\,dy = x^2\,dx.$$

Integrate:

$$y + \frac{y^3}{3} = \frac{x^3}{3} + C.$$

This is a valid implicit general solution.

Common mistake. When you integrate $\frac{1}{y}$, the result is

$$\ln|y|,$$

not $\ln y$ without absolute value.

8. 7.7 Particular Solutions Using Initial Conditions

Core idea

Use the initial condition to determine the constant in the general solution.

Worked example. Find the particular solution of

$$\frac{dy}{dx} = 3xy, \qquad y(0) = 5.$$

From separation of variables, the general solution is

$$y = Ce^{\frac{3x^2}{2}}.$$

Use $y(0) = 5$:

$$5 = Ce^0 = C.$$

Therefore,

$$y = 5e^{\frac{3x^2}{2}}.$$

Worked example. Solve the initial-value problem

$$\frac{dy}{dx} = 2x(1+y), \qquad y(0) = 0.$$

Separate variables:

$$\frac{1}{1+y}\,dy = 2x\,dx.$$

Integrate:

$$\ln|1+y| = x^2 + C.$$

Exponentiate:

$$1 + y = Ce^{x^2}.$$

Apply $y(0) = 0$:

$$1 = Ce^0 = C.$$

Hence,

$$y = e^{x^2} - 1.$$

9. 7.8 Exponential Models

Core idea

When the rate is proportional to the current amount, the solution is exponential.

Key idea. If

$$\frac{dy}{dt} = ky,$$

then

$$y = Ce^{kt}.$$

If $k > 0$, the model shows exponential growth. If $k < 0$, it shows exponential decay.

Worked example. A bacteria culture satisfies

$$\frac{dP}{dt} = 0.6P, \qquad P(0) = 200.$$

Find $P(t)$.

The solution has the form

$$P = Ce^{0.6t}.$$

Use $P(0) = 200$:

$$200 = Ce^0 = C.$$

Thus,

$$P(t) = 200e^{0.6t}.$$

Worked example. A radioactive sample satisfies

$$\frac{dM}{dt} = -0.12M, \qquad M(0) = 80.$$

Find $M(t)$.

This is exponential decay, so

$$M(t) = 80e^{-0.12t}.$$

10. 7.9 Logistic Models (BC)

Core idea

Logistic growth increases at first, then slows down as the amount approaches the carrying capacity.

Key idea. Standard logistic model:

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right)$$

where L is the carrying capacity.

Important facts:

$$P = 0 \quad \text{and} \quad P = L$$

are equilibrium solutions, and the growth rate is greatest at

$$P = \frac{L}{2}.$$

Worked example. For

$$\frac{dP}{dt} = 0.3P\left(1 - \frac{P}{500}\right),$$

find the carrying capacity and the value of P at which growth is fastest.

The carrying capacity is

$$L = 500.$$

The growth is fastest at

$$P = \frac{500}{2} = 250.$$

Worked example. For

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{1200}\right),$$

determine whether the population is increasing or decreasing when $P = 1500$.

Compute the sign of the factor:

$$1 - \frac{1500}{1200} = 1 - 1.25 = -0.25.$$

Thus,

$$\frac{dP}{dt} = k(1500)(-0.25) < 0,$$

so the population is decreasing.

11. Method Selection Guide

Key idea. Use this quick guide during practice and on the exam:

- Words only $\to$ build a model.
- Proposed function $\to$ verify by differentiating and substituting.
- Slope field picture $\to$ reason from signs and horizontal rows/vertical columns.
- Numerical estimate after a small step size $\to$ Euler's Method.
- Equation can be separated into all y's and all x's $\to$ separation of variables.
- Form $y' = ky \to$ exponential model.
- Form $P' = kP(1 - P/L) \to$ logistic model.

12. High-Yield Formula Sheet

$$\frac{dy}{dt} = ky \quad \Rightarrow \quad y = Ce^{kt}$$

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right)$$

$$y_{n+1} = y_n + h\, f(x_n, y_n)$$

$$\frac{dy}{dx} = g(x)h(y) \quad \Rightarrow \quad \frac{1}{h(y)}\, dy = g(x)\, dx$$

$$P = 0,\ P = L \quad \text{(equilibrium values in a logistic model)}$$

$$P = \frac{L}{2} \quad \text{(fastest growth in a logistic model)}$$

13. Mixed Practice with Answers

Problem 1

A population satisfies

$$\frac{dP}{dt} = 0.4P, \qquad P(0) = 3.$$

Find the particular solution.

Worked example. The solution has the form

$$P = Ce^{0.4t}.$$

Using $P(0) = 3$, we get $C = 3$. So

$$P(t) = 3e^{0.4t}.$$

Problem 2

Use Euler's Method with step size 0.1 to estimate $P(0.1)$ if

$$\frac{dP}{dt} = 0.4P, \qquad P(0) = 3.$$

Worked example. At $t = 0$, the slope is

$$P'(0) = 0.4(3) = 1.2.$$

So

$$P(0.1) \approx 3 + 0.1(1.2) = 3.12.$$

Problem 3

Solve the initial-value problem

$$\frac{dy}{dx} = \frac{x}{y}, \qquad y(1) = 2.$$

Worked example. Separate variables:

$$y\,dy = x\,dx.$$

Integrate:

$$\frac{y^2}{2} = \frac{x^2}{2} + C.$$

Multiply by 2:

$$y^2 = x^2 + C.$$

Use $y(1) = 2$:

$$4 = 1 + C \Rightarrow C = 3.$$

Thus,

$$y = \sqrt{x^2 + 3}$$

using the positive branch because $y(1) = 2$.

Problem 4

For

$$\frac{dP}{dt} = 0.2P\left(1 - \frac{P}{800}\right),$$

find the equilibrium solutions and the population at which growth is fastest.

Worked example. Set the derivative equal to zero:

$$0.2P\left(1 - \frac{P}{800}\right) = 0.$$

So the equilibrium solutions are

$$P = 0 \quad \text{and} \quad P = 800.$$

The growth is fastest at

$$P = \frac{800}{2} = 400.$$

14. Final Checklist for Unit 7

Key idea. Before the AP exam, you should be able to do all of the following quickly:

1. Translate a verbal description into a differential equation.
2. Verify whether a proposed function is a solution.
3. Read a slope field and identify increasing, decreasing, and equilibrium behavior.
4. Use Euler's Method for one-step and multi-step approximations.
5. Solve separable equations.

6. Use an initial condition to find a particular solution.
7. Recognize exponential growth and decay.
8. Recognize logistic growth, identify the carrying capacity, and know that the fastest growth occurs at $L/2$.

End of Unit 7 Review Book